COMP5331 Knowledge Discovery in Databases (Fall Semester 2026)
Homework 1
Deadline: 9 Oct, 2026 1:30pm
(Please hand in Canvas.)
Full Mark: 100 Marks

Coupon Instructions:

1. You can use a coupon to waive any question you want and obtain full marks for this question.
2. You can waive at most one question in each assignment.
3. You can also answer the question you will waive. We will also mark it but will give full marks to this question.
4. The coupon is non-transferrable. That is, the coupon with a unique ID can be used only by the student who obtained it in class.
5. Please scan the coupon (hardcopy) as an electronic copy
6. Please place this electronic copy of the coupon to your the submitted assignment (softcopy) on the first page
7. Please write down/type the question no. you want to waive on the coupon.

Q1 [20 Marks]

(a) In general, we have a number of customers. For illustration, we are given two customers, namely X and Y. The following shows 5 transactions for these two customers. Each transaction contains three kinds of information: (1) customer ID (e.g., X and Y), (2) the time that this transaction occurred, and (3) all the items involved in this transaction.

Customer X, time 1, items A, B, C
Customer Y, time 2, items A, F
Customer X, time 3, items D, E
Customer X, time 4, item G
Customer Y, time 5, items D, E, G

For example, the first transaction corresponds to that customer X bought item A, item B and item C at time 1, while the last transaction corresponds to that customer Y bought item D, item E and item G at time 5.

A sequence is defined to be a series of itemsets in form of $<S_1, S_2, S_3, \ldots, S_m>$ where S_i is an itemset for $i = 1, 2, \ldots, m$. The above transactions can be transformed into two sequences as follows.

X: <{A, B, C}, {D, E}, {G}>
Y: <{A, F}, {D, E, G}>

After this transformation, each customer is associated with a sequence.

Given a sequence S in form of $<S_1, S_2, S_3, \ldots, S_m>$ and another sequence S' in form of $<S_1', S_2', S_3', \ldots, S_n'>$, S is said to be a subsequence of S' if $m \leq n$ and there exist m integers, namely $i_1, i_2, \ldots, i_m$, such that (1) $1 \leq i_1 < i_2 < \ldots < i_m \leq n$, and (2) $S_j \subseteq S_{i_j}'$ for $j = 1, 2, \ldots, m$. If S is a subsequence of S', then S' is defined to be a super-sequence of S.
The support of a sequence S is defined to be the total number of customers which sequences are super-sequences of S.

Given a positive integer k, a sequence in form of $<S_1, S_2, S_3, \ldots, S_m>$ is said to be a k-sequence if $\sum_{i=1}^{m} |S_i|=k$.

Can the Apriori algorithm be adapted to mining all k-sequences with support at least 2 where k = 2, 3, 4, …. ? If yes, please write down the proposed method using the concept of the Apriori algorithm and illustrate your algorithm with the above example. If no, please explain the reason.

(b) We want to study the same problem setting described in (a). However, each customer is associated to one binary attribute called "Rich" to indicate whether this customer is rich or not. There are only 2 possible values in this attribute, namely "Yes" and "No". In our example, customer X could have "Yes" in attribute "Rich" and customer Y could have "No" in attribute "Rich".

Given a k-sequence S and a value v in attribute "Rich", the support of a sequence S with respect to value v is defined to be the total number of customers which sequences are super-sequences of S and are associated with value v in attribute "Rich". The important ratio of S is defined to be the support of S with respect to value "Yes" divided by the support of S with respect to value "No".

Can the Apriori algorithm be adapted to mining all k-sequences with important ratio at least 2 and the support at least 1 where k = 2, 3, 4, ….? If yes, please write down the proposed method using the concept of the Apriori algorithm and illustrate your algorithm with the above example. If no, please explain the reason.

Note that when we compute the important ratio, if we encounter a division of a non-zero number by zero, we could regard it as a positive infinity value.

Q2 [20 Marks]

An itemset Y is a *proper super-itemset* of X if $X \subset Y$.
An itemset X is said to be *closed* if there exists no proper super-itemset Y such that Y has the same support count as X.
Consider the following transactions. Each row corresponds to a transaction where 1 corresponds to a presence of an item and 0 corresponds to an absence.

A	B	C	D
1	0	0	1
0	1	0	0
1	0	1	1
1	0	1	1

(a) Please adapt algorithm FP-growth with the use of the FP-tree to find all closed itemset which support is at least the K-th greatest support among the multi-set containing the supports of all closed itemsets where K is a positive integer.
   (i) Please write down how to adapt algorithm FP-growth.
   (ii) Please illustrate the algorithm in (i) with the above example when K is set to 2.
   (iii) Repeat (ii) when K is set to 3.

(b) What are the advantages and the disadvantages of using these closed itemsets compared with traditional frequent itemsets you leant in class?

Q3 [20 Marks]

Consider the following eight two-dimensional data points:

x_1:(15, 10), x_2: (3, 10), x_3: (15, 12), x_4: (3, 14), x_5: (18, 13), x_6: (1, 7), x_7: (10, 1), x_8: (10, 30)

Consider algorithm k-means.

(a) Please answer the following questions. You are required to show the information about each final cluster (including the mean of the cluster and all data points in this cluster). You can consider writing a program for this part but you are not required to submit the program.
   (i) If k = 2 and the initial means are (10, 1) and (10, 30), what is the output of the algorithm?
   (ii) If k = 2 and the initial means are (3, 10) and (15, 10), what is the output of the algorithm?
   (iii) If k = 3 and the initial means are (10, 1), (10, 30) and (3, 10), what is the output of the algorithm?
   (iv) If k = 4 and the initial means are (10, 1), (10, 30), (3, 10) and (15, 10), what is the output of the algorithm?

(b) What are the advantages and the disadvantages of algorithm k-means?
   For each disadvantage, please also give a suggestion to enhance algorithm k-means.

Q4 [20 Marks]

Consider eight data points.

The following matrix shows the pairwise distances between any two points.

	1	2	3	4	5	6	7	8
1	0							
2	11	0						
3	5	13	0					
4	12	2	14	0				
5	7	17	1	18	0			
6	13	4	15	5	20	0		
7	9	15	12	16	15	19	0	
8	11	20	12	21	17	22	30	0

(a) Please use the agglomerative approach to group these points with distance group average linkage. Draw the corresponding dendrogram for the clustering. You are required to specify the distance metric in the dendrogram.

(b) Suppose that we want to find 4 clusters. According to the dendrogram in (a), please state the 4 clusters. For each cluster, please include all data points involved.

(c) (i) What is the greatest possible number of data points in a cluster containing data 1 and data 5?
   (ii) What is the smallest possible number of data points in a cluster containing data 1 and data 5?

(d) Suppose that data points satisfy the triangle inequality. That is, for any three data points, a, b and c, we have $|a, b| + |b, c| \geq |a, c|$ where $|a, b|$ denotes the pairwise distance between a and b, and $|b, c|$ and $|a, c|$ have similar meanings. Does the triangle inequality enhance the agglomerative approach? If yes, please elaborate it. If no, please give the reason.

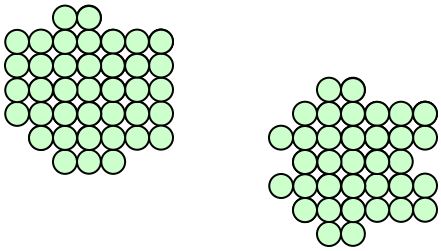
Q5 [20 Marks]

(a) One disadvantage of method k-means is that k (i.e., the number of clusters) should be pre-determined. One may suggest the following method to determine parameter k.

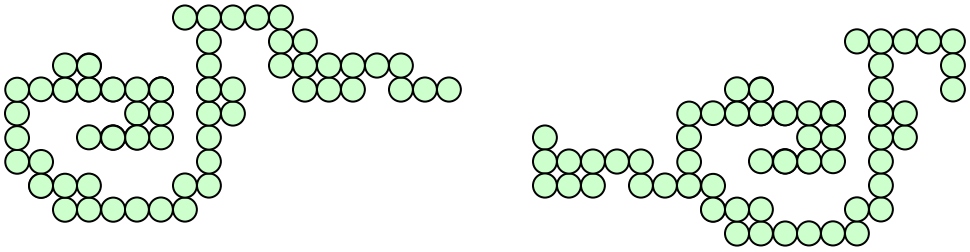
- **Step 1:** Set variable e_0 to ∞
- **Step 2:** Set k to 1 initially
- **Step 3:** Run the original k-means method and obtain k cluster centers (or means)
- **Step 4:** Set variable e_k to the sum of the distances between points and their closest cluster centers (according to the k cluster centers found).
- **Step 5:** If e_k converges (i.e., $(e_{k-1} - e_k)$ is equal to 0 or an extremely small number), then return k. Otherwise, increment k by 1 and repeat Step 3 to Step 4.

Can the above method determine a good value for k (i.e., the number of clusters)? Please explain.
If your answer is no, please also give an algorithm to determine a good value for k and explain why it is better than the above method.

(b) In a typical setting for k-menas, we calculate the distance between a point and a cluster by computing the Euclidean distance between a point and the center (or mean) of the cluster.
However, this computation method can only capture "circular" clusters as shown below.



In some cases, clusters are not "circular". Instead, they may appear as lines as shown below.



Please give a method or a distance function to capture both the "circular" clusters and the "appear-as-line" clusters.